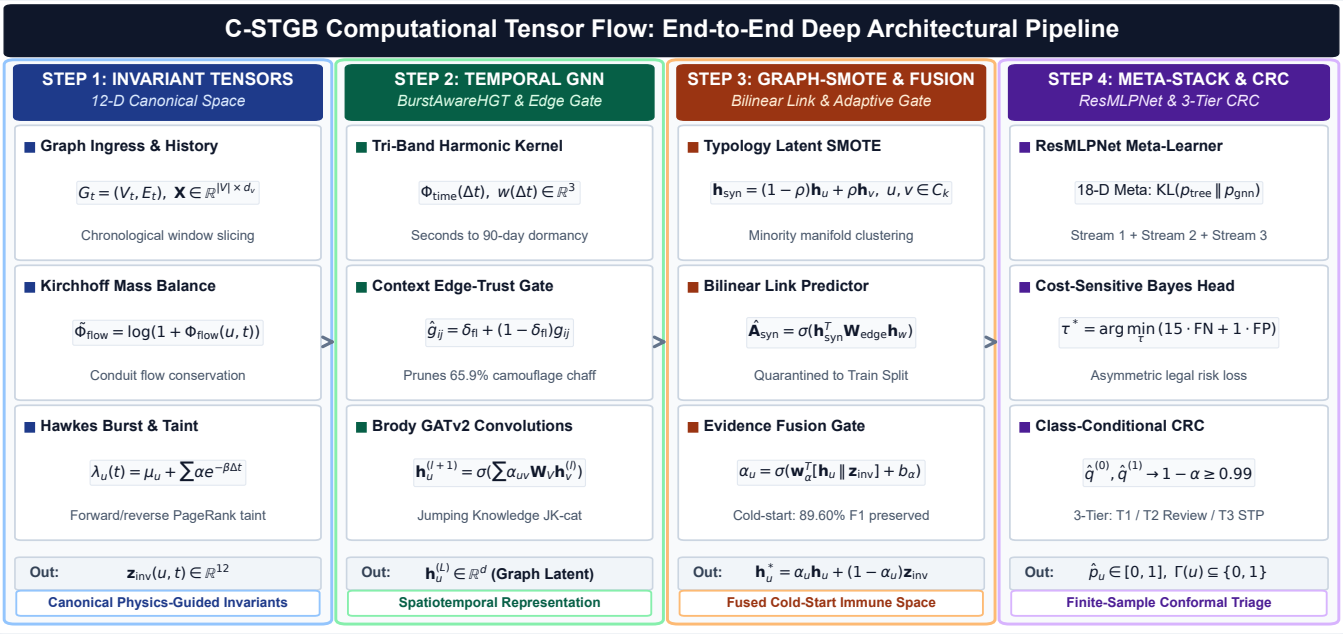

C-STGB Computational Tensor Flow: End-to-End Deep Architectural Pipeline
STEP 1: INVARIANT TENSORS
12-D Canonical Space
Graph Ingress & History
$G_t = (V_t, E_t),\ \mathbf{X} \in \mathbb{R}^{|V| \times d_v}$
Chronological window slicing
Kirchhoff Mass Balance
$\tilde{\Phi}_{\text{flow}} = \log(1 + \Phi_{\text{flow}}(u, t))$
Conduit flow conservation
Hawkes Burst & Taint
$\lambda_u(t) = \mu_u + \sum \alpha e^{-\beta \Delta t}$
Forward/reverse PageRank taint
Out: $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$
Canonical Physics-Guided Invariants
STEP 2: TEMPORAL GNN
BurstAwareHGT & Edge Gate
Tri-Band Harmonic Kernel
$\Phi_{\text{time}}(\Delta t),\ w(\Delta t) \in \mathbb{R}^3$
Seconds to 90-day dormancy
Context Edge-Trust Gate
$\hat{g}_{ij} = \delta_{\text{fl}} + (1 - \delta_{\text{fl}}) g_{ij}$
Prunes 65.9% camouflage chaff
Brody GATv2 Convolutions
$\mathbf{h}_u^{(l+1)} = \sigma(\sum \alpha_{uv} \mathbf{W}_V \mathbf{h}_v^{(l)})$
Jumping Knowledge JK-cat
Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Graph Latent)
Spatiotemporal Representation
STEP 3: GRAPH-SMOTE & FUSION
Bilinear Link & Adaptive Gate
Typology Latent SMOTE
$\mathbf{h}_{\text{syn}} = (1 - \rho)\mathbf{h}_u + \rho \mathbf{h}_v,\ u, v \in C_k$
Minority manifold clustering
Bilinear Link Predictor
$\hat{\mathbf{A}}_{\text{syn}} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$
Quarantined to Train Split
Evidence Fusion Gate
$\alpha_u = \sigma(\mathbf{w}_\alpha^T [\mathbf{h}_u \,\|\, \mathbf{z}_{\text{inv}}] + b_\alpha)$
Cold-start: 89.60% F1 preserved
Out: $\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u) \mathbf{z}_{\text{inv}}$
Fused Cold-Start Immune Space
STEP 4: META-STACK & CRC
ResMLPNet & 3-Tier CRC
ResMLPNet Meta-Learner
18-D Meta: $\text{KL}(p_{\text{tree}} \,\|\, p_{\text{gnn}})$
Stream 1 + Stream 2 + Stream 3
Cost-Sensitive Bayes Head
$\tau^* = \arg\min_\tau (15 \cdot \text{FN} + 1 \cdot \text{FP})$
Asymmetric legal risk loss
Class-Conditional CRC
$\hat{q}^{(0)}, \hat{q}^{(1)} \to 1 - \alpha \geq 0.99$
3-Tier: T1 / T2 Review / T3 STP
Out: $\hat{p}_u \in [0, 1],\ \Gamma(u) \subseteq \{0, 1\}$
Finite-Sample Conformal Triage